Software Requirements Specification for Software Engineering: subtitle describing software

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Contents

Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

Introduction

The software requirements specification (SRS) document describes the functional and non-functional requirements, scope and functionality of the application. It formally outlines key measures relevant to all the different stages of the software development lifecycle. This document is constructed using the Volere Software Requirements Specification template [Robertson and Robertson, 2012].

1 Purpose of the Project

1.1 User Business

According to the Global Burden of Diseases, Injuries and Risk Factors study performed in 2019,individuals that would benefit from physical rehabilitation at least once in their lifetime is upwards of 2.41 billion globally [Cieza et al., 2021]. Those with access to a physiotherapist experienced a disconnect with performing a required movement with proper range of motion (ROM) and correct form [Faber et al., 2015]. While a physiotherapist can advise these individuals during their assessments and proceeding follow-up appointments, the efficacy of rehabilitation depends heavily on the individual's correct performance of the exercise. In turn, this creates a need for a tool that can ensure users correctly perform the exercise without supervision. This project aims to develop a tool that can provide feedback and corrections for the prescribed physical rehabilitation exercise.

1.2 Goals of the Project

Insert your content here.

2 Naming Conventions and Terminology

Insert your content here.

3 Stakeholders

The stakeholders for this project are parties with a vested interest in the end product. They will be consulted on their requirements and opinions throughout the development process. Their impact will be foundational for correctness of the product. This section outlines the various types and levels of stakeholders, the personas, user participation (towards application development) and any users that will be responsible for the product's maintenance.

3.1 Client

There are two categories of clients that are the target audience for this product. The first category of end users is the patients. The application is being designed for patients who are undergoing physiotherapy treatment and have been given a home exercise program to aid in their rehabilitation. For the purpose of this system, the exercise and the joint selected was recommended by one of our stakeholders. The knee and a straight leg raise were chosen.

The second category of the clients are the physiotherapists who are treating the users on the application. It will integrate the physiotherapist's patient list and their corresponding performance statistics. The application allows physiotherapists to gain insight into how their patients' are progressing. Furthermore, it will enable the assessment of the suitability of the prescribed exercise, as well as gauging whether any additional modifications are required (number of repetitions, degree of complexity etc.).

3.2 Customer

The application is being developed for a Canadian market with its governing standards and regulations in mind. Although this version of the application does not have any customers, it can prospectively be purchased by any healthcare provider and be incorporated into their patient care structure.

3.3 Other Stakeholders

The primary and secondary stakeholders are described above (clients and customers).

Other stakeholders may include regulatory authorities or other healthcare specialists, such as physiatrists or massage therapists. The regulatory authorities will be allowed access to conduct audits and ensure that any disclosed patient information is being handled in an ethical manner. In addition, they will verify the accuracy and quality of care being delivered. Other healthcare providers may benefit from any insights or information that is being provided, which may be used to tailor their own care of the patient. Lastly, support workers may use this application to assist their clients with their home exercises.

3.4 Hands-On Users of the Project

Patients
User role: Record corresponding exercise for application to analyze, adjust exercise form based on application feedback and correction, gain insight from performance metrics provided by the application.
Subject matter experience: Aware of how to perform exercise using the instructions provided by the physiotherapist (eg. a worksheet detailing the steps required for the exercise).
Technological experience: Know how to operate the smart device and use its camera to record their performance of the exercise.
Attitude towards technology: Open to learning if needed.
Physical abilities/disabilities: Able to use a smart device and/or perform the prescribed exercise.
Physiotherapists
User role: Access patient statistics to assess suitability of exercise, difficulty and whether the prescribed exercise is effective.
Subject matter experience: Formal education on the topic and have knowledge of general and post operative care.
Technological experience: Able to effectively use the various functionalities of the application.
Attitude towards technology: Varied.
Education: Officially licensed.

3.5 Personas

Name: Scotty Fye
Age: 72
Occupation: Retired (Former teacher)

Location: Toronto, Canada
Status: Married, with children

"I had knee surgery a few weeks ago because my knee pain was unbearable. As suggested by my doctor, I had knee replacement surgery. Now, I am following some prescribed exercises but I'm worried that I am doing them wrong and possibly slowing down recovery. On top of that, I have a vacation with my family and want to be the 'fun grandpa'".

- Goals:
  - To gain mobility and strength in his knee for an upcoming vacation and keep up with his active grandchildren.
  - He wants to understand how to both manage pain and recover well.
  - See how he is performing on a weekly basis.
  - Have guidance for performing these exercises alone to ensure he is performing exercises correctly.
- Fear:
  - Worried about doing something wrong which might cause pain and hinder his recovery.
  - Prescribed exercises are manageable but is unsure if he should push through discomfort.
  - "Should I push through stiffness or should I stop?"
  - "Am I bending my knee enough? Not enough?"

Name: Maya Thompson
Age: 21
Occupation: Student
Location: Hamilton, Ontario

“I was in a car accident which shattered my knee. I had to get a full knee replacement in order to continue walking. I was recently discharged from a rehabilitation facility where I started learning how to use my knee and walk again. My physiotherapist enrolled me as their patient on this application to keep up to date on my progress towards recovery.”

- Goal:
  - To gain sufficient use of her knee to walk and perform activities of daily living.
  - To monitor her own progress and use it as a motivating factor for pushing herself.
  - To gain enough strength in her muscles without having to further exacerbate her pain.
- Fear:
  - Her muscles need to relearn how to function and she is afraid if she performs an exercise incorrectly, the incorrect muscles will learn to compensate during the movement.
  - ‘Am I using the correct muscles? Is this the muscle that I need to consciously activate?’
  - Her pain might be aggravated to an intolerable degree.
  - ‘What if my pain does not start out as much but becomes unbearable because I did not perform the exercise correctly?’

3.6 Priorities Assigned to Users

Key users: Patients undergoing post operative physiotherapy have highest priority in the system, physiotherapists using the system to check-in and comment on patient exercises have similar priority.

3.7 User Participation

Users (Patients)

- User creates an account and set up their profile and may sign into this account with valid credentials.
- User requests analytical information based on their past exercise inputs and results.
- User receives live form recommendations from the system by providing a video-feed of themselves performing the selected activity.
- User requests an overview of their exercise after successfully completing a video-feed upload, or looks back at old overviews.
- User creates comments on their own exercise overviews to keep personal notes and information regarding exercise session.
- User receives comments from the physiotherapist account linked to theirs.

Users (Physiotherapists)

- User creates an account and sets up their profile, and may sign into this account with valid credentials.
- User requests a list of all patient accounts connected to their account.
- User receives overviews of patient's exercises to confirm exercise completion and progress on recovery.
- User creates comments on patient's exercise overviews to assist the patient with exercise performance, and/or to provide modifications.

3.8 Maintenance Users and Service Technicians

Maintenance and service on the system will be performed by the current development team. Maintenance and service should be minimized with a thorough test suite to ensure all functionalities of the system are operational upon any changes or updates.

4 Mandated Constraints

4.1 Solution Constraints

Insert your content here.

4.2 Implementation Environment of the Current System

Insert your content here.

4.3 Partner or Collaborative Applications

Insert your content here.

4.4 Off-the-Shelf Software

Insert your content here.

4.5 Anticipated Workplace Environment

Insert your content here.

4.6 Schedule Constraints

Insert your content here.

4.7 Budget Constraints

Insert your content here.

4.8 Enterprise Constraints

Insert your content here.

5 Naming Conventions and Terminology

5.1 Glossary of All Terms, Including Acronyms, Used by Stakeholders involved in the Project

Insert your content here.

6 Relevant Facts And Assumptions

6.1 Relevant Facts

Relevant facts for this project are that about a third of patients needing physiotherapy can have improved efficacy due to their exertion and form factors. This leads to slower recovery or future injury later down the line. In some instances, this can lead to a permanent decrease in an affected joint's range of motion and strength. Additionally, there is a gap from initially being prescribed the exercise and the accountability from doing exercises on the patients own time.

The primary target of this system is for patients of physiotherapists and physiotherapists.

6.2 Business Rules

Some business rules relevant to include that there will be a distinct user interface for general users of the application and a different user interface for physiotherapists. A patient must be directed to use the application from the physiotherapist and will have exercises assigned to them on the system. Similarly, a physiotherapist can view and manage their patients but cannot access other patients outside their care.

A prescribed exercise chosen from the physiotherapist can have certain parameters and limitations, for example: sets, reps, frequency, etc. As well, the patient will be prompted after an exercise to give their perceived exertion as a result of the exercise from a sale of 1-10.

Furthermore, any data will not be shared without any explicit consent and stored securely.

6.3 Assumptions

Insert your content here.

7 The Scope of the Work

7.1 The Current Situation

Insert your content here.

7.2 The Context of the Work

Insert your content here.

7.3 Work Partitioning

Insert your content here.

7.4 Specifying a Business Use Case (BUC)

Insert your content here.

8 Business Data Model and Data Dictionary

8.1 Business Data Model

Insert your content here.

8.2 Data Dictionary

Insert your content here.

9 The Scope of the Product

9.1 Product Boundary

Insert your content here.

9.2 Product Use Case Table

Insert your content here.

9.3 Individual Product Use Cases (PUC's)

Insert your content here.

10 Functional Requirements

10.1 Functional Requirements

Insert your content here.

11 Look and Feel Requirements

11.1 Appearance Requirements

Insert your content here.

11.2 Style Requirements

Insert your content here.

12 Usability and Humanity Requirements

12.1 Ease of Use Requirements

Insert your content here.

12.2 Personalization and Internationalization Requirements

Insert your content here.

12.3 Learning Requirements

Insert your content here.

12.4 Understandability and Politeness Requirements

Insert your content here.

12.5 Accessibility Requirements

Insert your content here.

13 Performance Requirements

13.1 Speed and Latency Requirements

Insert your content here.

13.2 Safety-Critical Requirements

Insert your content here.

13.3 Precision or Accuracy Requirements

Insert your content here.

13.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

13.5 Capacity Requirements

Insert your content here.

13.6 Scalability or Extensibility Requirements

Insert your content here.

13.7 Longevity Requirements

Insert your content here.

14 Operational and Environmental Requirements

14.1 Expected Physical Environment

Insert your content here.

14.2 Wider Environment Requirements

Insert your content here.

14.3 Requirements for Interfacing with Adjacent Systems

Insert your content here.

14.4 Productization Requirements

Insert your content here.

14.5 Release Requirements

Insert your content here.

15 Maintainability and Support Requirements

15.1 Maintenance Requirements

Insert your content here.

15.2 Supportability Requirements

Insert your content here.

15.3 Adaptability Requirements

Insert your content here.

16 Security Requirements

16.1 Access Requirements

Insert your content here.

16.2 Integrity Requirements

Insert your content here.

16.3 Privacy Requirements

Insert your content here.

16.4 Audit Requirements

Insert your content here.

16.5 Immunity Requirements

Insert your content here.

17 Cultural Requirements

17.1 Cultural Requirements

Insert your content here.

18 Compliance Requirements

18.1 Legal Requirements

Insert your content here.

18.2 Standards Compliance Requirements

Insert your content here.

19 Traceability from Requirements to Use Cases

This section maps each of the requirements to one or more of the use cases listed and defined above in previous sections.

Table 1: Traceability from Requirements to Use Cases

Requirement	**Use Case(s)**
FR1	PUC3, PUC10
FR2	PUC1
FR3	PUC3
FR4	PUC10
FR5	PUC8
FR6	PUC10
FR7	PUC5
FR8	PUC7
FR9	PUC9
FR10	PUC4, PUC8
FR11	PUC11
FR12	PUC11
FR13	PUC11, PUC12
FR14	PUC6
FR15	PUC7
FR16	PUC7
FR17	PUC8, PUC10
FR18	PUC10
FR19	PUC9
FR20	PUC10
FR21	PUC13
FR22	PUC10, PUC13
FR23	PUC1, PUC13
FR24	PUC13

Continued on next page

Table 1 – continued from previous page

Requirement	Use Case(s)
FR25	PUC8, PUC10
LFR-AP1	All PUCs
LFR-ST1	PUC4, PUC6, PUC8
UHR-EOU1	All PUCs
UHR-EOU2	PUC6
UHR-LRN1	PUC5, PUC8
UHR-UPL1	PUC6
UHR-AR1	PUC5, PUC6, PUC7
PR-SL1	PUC5
PR-SL2	PUC6
PR-SCR1	PUC6, PUC7
PR-PAR1	PUC11
PR-PAR2	PUC11, PUC12
PR-RFT1	PUC5
PR-RFT2	PUC5
PR-RFT3	All PUCs
PR-CR1	All PUCs
PR-SER1	PUC11, PUC12
OER-EP1	PUC5, PUC11
OER-WE1	All PUCs
OER-WE2	All PUCs
OER-PR1	PUC1, PUC3, PUC5, PUC6, PUC8, PUC10
OER-RL1	PUC5, PUC11, PUC12
MS-MNT1	PUC13
MS-AD1	PUC5, PUC8
SR-AR1	PUC1
SR-AR2	PUC3
SR-AR3	PUC2
SR-IR1	PUC13
SR-PR1	PUC1, PUC13
SR-PR2	PUC13

20 Open Issues

This section discusses the open issues in this project. In detail, we will highlight the primary uncertainties and open questions that are present throughout the process of developing our physiotherapy app. Ultimately, these issues represent the challenges we may face throughout the development of our app. Addressing these issues will aid us in ensuring the app is efficient, accurate and reliable. Listed below are some potential open issues in this project we have come across, as well as how we plan to address these issues:

- **Exercise Detection Accuracy:** Ensuring the app correctly analyzes and identifies the movements the patient undergoes is always going to be a constant challenge. Inaccuracies in the detection could result in false feedback which may prohibit recovery time. To address this, we will be undergoing further refinement of our detection methods and testing with a diverse sample size of individuals to aid in the accuracy of our movement detection to provide the most accurate feedback.
- **Data Privacy:** The app will collect and store sensitive client health information which is something that needs to be securely stored. We will be making sure information is securely stored, transmission is encrypted and there are strict access controls in place to have that trust exist with our clients.
- **Maintaining Patient Engagement:** Maintaining patient engagement is something that is going to be a struggle, as some patients may view this as extra work on their end. Our goal is to hopefully have this app be viewed as a convenient aid that reduces the need for frequent in person physiotherapy visits, making recovery more so accessible and less time-consuming for the clients.

21 Off-the-Shelf Solutions

21.1 Ready-Made Products

Insert your content here.

21.2 Reusable Components

Insert your content here.

21.3 Products That Can Be Copied

Insert your content here.

22 New Problems

22.1 Effects on the Current Environment

1. **Increased Battery Usage of Smart Device:** The app's use of motion tracking and video processing could result in an increase of battery usage for a user's smart device. This could also limit usability for individuals who have older devices.

2. **Increased Storage Impact:** Since the app does involve uploading recording videos for analysis, this may result in an increase in storage used on the smart device. Individuals using the app who may have limited storage may experience a higher amount of data being used. This ultimately means that we need to ensure that due to the increase in storage usage, the quality is not lost, as we do not want that to affect the precision of our analysis in any way, shape or form.

3. **Requirement of stable network access:** The app does rely on a stable network connection to upload exercise videos for analysis. Areas with weaker connections may lead to individuals dealing with more delays and thus, limiting their overall usability.

22.2 Effects on the Installed Systems

1. It may be hard for existing clinic or hospital systems to easily support integration with our app. There may be incompatibilities between the varying software platforms used, so this may slow down the process of integration or require more effort and money to aid in integrating it more smoothly.

2. Different systems may store data differently so this is something we also have to be aware of as it could create inconsistencies when syncing the medical records.

3. When integrating the app to healthcare networks, policies and authentication layers should be carefully managed because if proper security is not in place, unauthorized users may be able to access patient data, which is extremely unsafe.

22.3 Potential User Problems

1. **Learning Curve:** Some users may not be the most familiar with using technology and may find it a bit more difficult to use this app which ultimately may lead to a decrease in user satisfaction and engagement.

2. **Privacy Issues and Comfortability with Video Recordings:** Some users may not be comfortable with the process of recording themselves and uploading their videos to the platform, which could lead to a decrease in engagement.

3. **Preference of in-person physiotherapy guidance:** Some users may prefer in person appointments with their physiotherapists to receive direct feedback, which may lead to a decrease in user engagement overall.

4. **Limitations of Recording Videos:** Some users may find it difficult as a result of their limited mobility to be able to record and upload the videos with precision by themselves without any additional support. This may lead to inaccurate computations and results.

5. **Misinterpretation of Feedback:** Users may not understand how to correctly translate or interpret the feedback given by the app, which could ultimately lead to a decrease in the effectiveness and overall productivity of the app.

22.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

1. **Limited access to technology:** Some users may be limited in terms of not owning smart devices with a functional camera or not having enough storage space to record and upload videos. This will ultimately limit the app's overall accessibility.

2. **Connectivity Issues:** Areas with weaker connections lead to individuals being more prone to experiencing delays and deal with uploads failing, which ultimately decreases the app's overall reliability and effectiveness.

22.5 Follow-Up Problems

1. **Regular Updates:** The app will consist of regular updates to make sure it is compatible with the latest operating systems, all the changes in the newest programming languages as well as ensure it abides by all of the standard in place.

2. **Data Management and Cleanup:** Large amounts of data will be stored, resulting in a lot of storage being used up, as mentioned previously. Potentially introducing some sort of periodic cleanup may need to be implemented to address this.

3. **On-going support may be required:** Users may require technical or instructional support when they are faced with issues while using the app specifically when recording, uploading or even trying to interpret the results. Potentially providing client support or a FAQ section may be necessary to address this.

23 Tasks

23.1 Project Planning

The project planning for this project will follow an agile development approach. We will follow the process of: planning, designing, developing, testing, deploying then reviewing. This will allow us to work towards manageable goals while testing and receiving feedback on our application.

Moreover some recurring tasks that we foresee happening will generally follow the flow below:

- Initial development of a goal for the milestone.
- Creation of issues on GitHub.
- Resource allocation and sprint execution.
- Testing and stakeholder review/feedback.
- Reflection on the previous sprint to see what could have gone better.

23.2 Planning of the Development Phases

Similar to section 9 of the development plan document, we seek to adhere to the project schedule.

- M1 (Weeks 3 to 4): Problem Statement, Proof of Concept (POC) Plan, Development Plan, Team Charter
- M2 (Weeks 5 to 6): Requirements Document (SRS) Revision 0, Hazard Analysis Revision 0, Design Document Revision -1
- M3 (Week 7): Verification & Validation (V&V) Plan Revision 0
- M4 (Weeks 8 to 9): Integrated Draft Document (Problem Statement, SRS, Hazard Analysis, V&V Plan), Peer Reviews (whole document), Update draft with feedback
- M5 (Weeks 10 to 12): Proof of Concept Demonstration (document + results), Design Document Revision 0, V&V Report Revision 0, Refinement of SRS, Hazard Analysis, and V&V Plan based on results and feedback
- M6 (End of Term): Final Demonstration - Revision 1, EXPO Demonstration, Final Documentation Revision 1 (including Problem Statement, Development Plan, POC Plan, Requirements Document, Hazard Analysis, Design Document, V&V Plan, V&V Report, Extra Documentation 1, Extra Documentation 2 and Source Code)

24 Migration to the New Product

24.1 Requirements for Migration to the New Product

Insert your content here.

24.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

25 Costs

Currently, the cost of this project is 0 dollars. However, the only cost at this moment is the team members' time and expertise in order to bring this project to fruition. All the tools that are currently being used to build the product are free to use and open source.

25.1 Metrics for Estimation

Financial costs are not a concern at this stage. However, circumstances may change and require re-evaluation of any cost metrics to include this new development. In that case, a cost-benefit analysis would be performed in order to determine the correct value.
The metrics listed below estimate the time and effort put forth by a team member:

- The *quality* of commits and pushes - this can be assessed by the commit message's clarity and descriptiveness.
- The *quantity* of contributions made to the project is an additional indicator of effort. However, it should be contextualized in terms of quality.
- The quality of other deliverables can be similarly analyzed.

As a team, the collective time and effort will be measured by:

- The number of goals that have been achieved.

- The amount of stretch goals that may have been successfully achieved.
- The quality of the final project deliverables, measured using deliverable grades and TA's feedback.
- Number of test cases that have been devised.
- Cohesiveness of the final product, which can be quantified by how functional the final application is.
- The amount of time spent on a project deliverable.

25.2 Estimation Approach

Current estimations are based on the previous and ongoing deliverable and may be inaccurate. They are subject to modification throughout the project timeline.

25.3 Cost Breakdown

Tools and Software:
The cost is 0 dollars since all libraries and additional software will be open source.
Testing Environment:
The projected test suite will be formulated in house and will have no cost.
Development effort:

- Documentation (generating output, editing): 120 hours per member
- Deliverable revision: 20 hours per member
- Team meetings: 50 hours per member
- Software development: 400 hours per member
- Software testing: 100 hours per member

Total: 690 hours per member

26 User Documentation and Training

26.1 User Documentation Requirements

Insert your content here.

26.2 Training Requirements

Insert your content here.

27 Waiting Room

Insert your content here.

28 Ideas for Solution

In this section , we will highlight the potential technical approaches we may be using to implement our physiotherapy app. The ideas for solution can be seen listed below:

- **Pose Detection:** We plan on using the OpenCV library to select and analyze specific frames of the video uploaded to pick and identify specific key knee joint points using pose estimation techniques. In terms of the calculations, we can use Python's built-in math module, which we can use to calculate the specific calculations necessary.

- **Feedback Charts:** To showcase graphs and progress summary charts, we can use Python's built-in matplotlib library to generate them for us. These can be later integrated to allow patients and physiotherapists the option to interpret the graphs and overall progress from the visuals provided.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

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